

Phosphofructokinase (PFK) Exercise

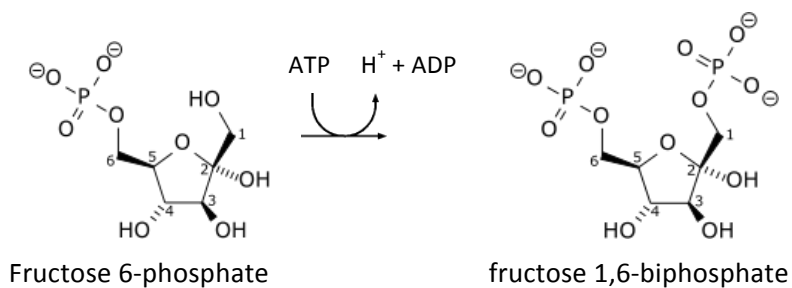
Learning Objectives

In this exercise, you will use StarBiochem, a protein 3D viewer, to explore the structure and function of the enzyme phosphofructokinase (PFK).

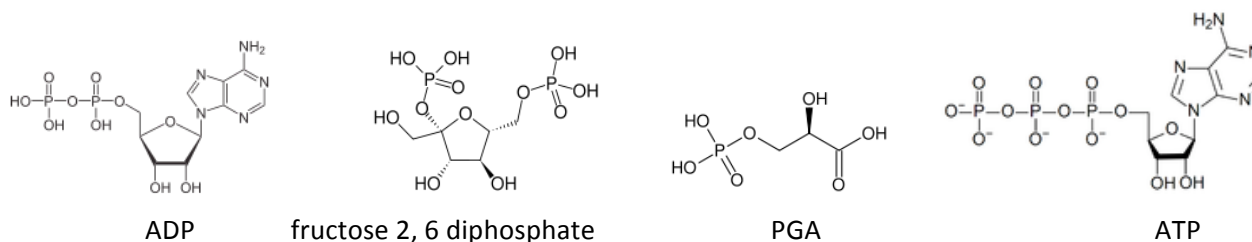
Background

Living organisms, both unicellular and multicellular, require energy to live and function. Energy is stored and provided in the form of adenosine triphosphate (ATP). ATP is produced by multiple cellular processes including glycolysis, as well as aerobic and anaerobic respiration.

PFK is the most important regulatory enzyme of glycolysis. It irreversibly catalyzes the rate-limiting step of glycolysis: the conversion of fructose 6-phosphate and ATP to fructose-1, 6-diphosphate and ADP.



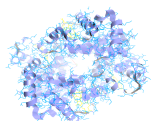
PFK exists as a homotetramer in bacteria and mammals. Each of the four monomers of PFK is composed of two similar domains. One domain has an ATP binding pocket. The other domain contains the substrate-binding site and the allosteric site. An allosteric site is a regulatory binding site that affects enzyme activity and is distinct from the substrate-binding site of an enzyme. Various activators and inhibitors regulate PFK. Examples of PFK activators are ADP, AMP and fructose 2, 6 diphosphate. Examples of PFK inhibitors are ATP, phosphoglycerate (PGA) and citrate.



Getting started with StarBiochem

- To begin using StarBiochem, please navigate to: <http://web.mit.edu/star/biochem>.
- Click on the **Start** button to launch the application.
- Click **Trust** when a prompt appears asking if you trust the certificate.
- Under **File**, click on **Open/Import** and select "2PFK" and click **Open**.

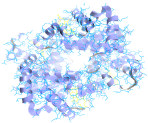
You are now viewing the structure of PFK (2PFK) with each bond in the protein drawn as a line ("bonds only" view). You will see two independent PFK structures within the 2PFK file. This is an artifact of how the structure was determined. Please focus on one of the two structures while answering the questions in this exercise.



Practice changing the viewpoint of this protein in the view window:

	Mac	PC
TO ROTATE	click and drag the mouse	left-click and drag the mouse
TO MOVE UP/DOWN RIGHT/LEFT	apple-click and drag the mouse	right-click and drag the mouse
TO ZOOM	option-click and drag the mouse	Alt-left-click and drag the mouse

Take a moment to look at the structure of PFK (2PFK) from various angles in this "bonds only" view. Before proceeding to answer the questions, you can review the basic structures and terms on the next page which you can refer to during this exercise.



PROTEIN STRUCTURE BASICS

Each protein has the following three levels of protein structure:

Primary structure

Lists the amino acids that make up a protein's sequence, but does not describe its shape.

Secondary structure

Describes regions of local folding that form a specific shape, like a helix, a sheet, or a coil.

Tertiary structure

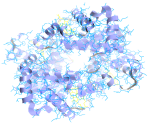
Describes the entire folded shape of a whole protein chain.

In addition, some proteins interact with themselves or with other proteins to form larger protein structures. How these proteins interact and fold to form a larger protein complex is termed **Quaternary structure**.

CHEMICAL STRUCTURES OF THE AMINO ACIDS

The 20 amino acids share a common backbone and are distinguished by different 'R' groups, highlighted in various colors below.

Non-polar									
	Glycine (Gly)	Alanine (Ala)	Valine (Val)	Leucine (Leu)	Isoleucine (Ile)	Methionine (Met)	Tryptophan (Trp)	Phenylalanine (Phe)	Proline (Pro)
Polar Uncharged									
	Serine (Ser)	Threonine (Thr)	Cysteine (Cys)	Tyrosine (Tyr)	Asparagine (Asn)	Glutamine (Gln)			
Polar Charged									
	Acidic/Negatively charged			Basic/Positively charged					
	Aspartic Acid (Asp)	Glutamic Acid (Glu)		Lysine (Lys)	Arginine (Arg)	Histidine (His)			



Protein Structure Questions

1 How many amino acids comprise PFK (2PFK)?

- Click on **Structure**.
- Click on **Primary** which shows the amino acids that are sequentially joined through peptide bonds to make the amino/polypeptide chain. The amino acids of each chain are highlighted by a specific color and can be distinguished from those of other chains.

Answer

2 How many monomers/chains do you see in the current view of PFK (2PFK)?

- To distinguish between the different monomers that make up PFK (2PFK), under **Structure** click on **Quaternary**.
- Click on **Chain**.

Answer

3 Does the current view of PFK (2PFK) represent the active form of this enzyme? Answer Yes / No and explain your answer.

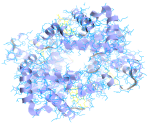
Answer

4 Let us highlight amino acids #195, #196 and #224 in the polypeptide chain.

- Under **Structure**, click on **Primary**.
- Select these amino acid residues by simultaneously clicking on **Control** and **Apple** key(Mac)/**right-click** (PC).
- The amino acids you select get highlighted in the crystal structure (white). For a better view you can go to **View Controls** and move the **Unselected** transparency slider to "0".
- Within the **Atoms** box click on **Draw** to visualize the atoms present. Each atom is color-coded: Carbon is grey, Nitrogen is blue, Oxygen is red and in this exercise Iron is orange.
- Alternatively move the **Backbone** slider to the left to visualize the atoms that are involved in bonding/interaction. You can zoom in the selected amino acids for a better view.

a) What is the **most likely** type of interaction between amino acids #196 and #224? Your choices are 'hydrogen bond', 'ionic bond', 'hydrophobic interaction', 'covalent bond', 'peptide bonds' or 'van der Waals forces'.

Answer



b) What is the most likely type of interaction between amino acids #195 and #196? Your choices are 'hydrogen bond', 'ionic bond', 'hydrophobic interaction', 'covalent bond', 'peptide bonds' or 'van der Waals forces'.

Answer

c) Draw amino acids #195 and #196 and circle the reactive group(s) or atoms that account for their interaction. Explain why you selected these groups (atoms).

Answer

d) Are these three amino acids part of the same or different polypeptide chain?

- Within **Structure**, go to **Quaternary** and click on **Chain**.
- Examine the color associated with these amino acids to determine if they are the same or different.

Answer

e) Highlight amino acids #241 and #261. What is the most likely type of interaction between these two amino acids? Your choices are 'hydrogen bond', 'ionic bond', 'hydrophobic interaction', 'covalent bond', 'peptide bonds' or 'van der Waals forces'.

Answer

5 Individual amino acids fold into different local secondary structures.

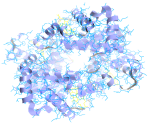
a) Name the different secondary structures observed in PFK (2PFK)?

- Under **Structure**, click on **Secondary**.
- Explore the different types of secondary structures found in PFK (2PFK) by individually clicking on the different choices that appear within the **Show Ribbons** box.

Answer

b) Please list the number of individual structures you observe for each of the secondary structures in PFK (2PFK). Focus on just one of the PFK monomers.

Answer



c) A cleft is found between the two domains that form the active site within each monomer. This cleft is formed by two sheets which are angled towards each other. **Identify the sheets within the structure that contribute to the formation of the active site.** Choose only one of the two independent PFK structures when answering this question.

Answer

d) We will now examine one of the sheets found within PFK (2PFK): S1C. **Is S1C an example of a parallel or an anti-parallel sheet?**

- Within **Secondary**, click on **Track Selection**.
- Within **Sheet Selection**, click on "S1C".
- Close **Structure** and click on **View Controls**.
- Move the **Unselected** transparency slider to the left until the rest of the protein falls into the background or completely disappears. Zoom in the selected helix also for a better view.
- To determine the location of the side chains move the **Sidechain** transparency slider to the left, while keeping the **Backbone** transparency slider to the right. Repeat this by moving the **Backbone** transparency slider to the left while keeping the **Sidechain** transparency slider to the right.
- Compare the orientation of the Oxygen atoms (**red**) within the C=O groups of each amino acid in one pleat of the sheet with the location of the same oxygen atoms in the next pleat.

Answer

6 Explore the tertiary structure of PFK (2PFK) and its relationship with the environment.

- Under **View**, click on **Reset molecule**.
- Click on **Structure** and select **Tertiary**.
- Within **Color by Residue** box click either individually or in combination the different options provided to answer the following questions.
- To improve visualization of the protein close **Structure**.
- Click **View Controls**.
- Within the **Atoms** box click on **Draw** and **Fill Space** to change the way 2PFK is viewed.

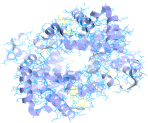
Why is the observed distribution of non-polar amino acids within PFK (2PFK) critical for the localization and function of this enzyme in the cytosol?

Answer

Protein Structure -> Function Questions

7 We will now explore PFK's catalytic activity. To do this we will compare the structure of PFK in different states.

- Click on **View** and choose **Reset Molecule**.
- Open two new windows of StarBiochem, while keeping the 2PFK window open.
- Click on the **Start** button to launch the application.



StarBiochem

Name _____

- Under **File** click on **Open/Import**.
- Click on “1PFK” and “6PFK” in each window.
- Click on **Open** for both windows.
- Compare the three PFK structures by going through the **Structure** menu.
- Click on **PDB Tree**. Compare the information provided in **PDB Tree**, in particular the ligands bound to the structures, with that provided in the “Background” section.

a) Which of these three structures represents the resting (ligand-unbound) state of PFK? Explain your answer.

Answer

b) Now compare the structure of the two remaining molecules that you did not select while answering the question above (7a). Which of the two structures represents the active state and which one represents the inactive state of PFK? Explain your choices.

Answer

Keywords:

Enzymes, regulatory/rate-limiting reactions, glycolysis, aerobic respiration, anaerobic respiration, catalytic reactions, reversible and irreversible reactions, allosteric modulators, allosteric activators, and allosteric inhibitors.

Thought Questions

- 1** Deficiency in PFK leads to Tarui's disease. This disorder is characterized by severe nausea, vomiting and muscle cramps. Patients with this genetic disorder are advised not to exercise vigorously. Explain why reduced physical activity can help these patients.
- 2** Activity of PFK influences the production of ATP both by aerobic and anaerobic respiration. Explain why this is so.
- 3** ATP is required for the reaction catalyzed by PFK. However, ATP is also an inhibitor of PFK. How could ATP perform two different regulatory effects on the same enzyme?